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# Curving resistance from wheel-rail interface

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## ABSTRACT

This paper uses vehicle system dynamics simulations to investigate curving resistance generated from the wheel-rail interface. More than 600 simulations were conducted using a freight wagon model to investigate the implications of curve radii, vehicle speeds and track superelevation for curving resistance, as well as the changes of curving resistance with wheelset Angle of Attack (AoA) and track cant deficiency. The results show that with the same curve radius, vehicle speeds and track superelevation can change the simulated curving resistance by 13–26%. Comparatively, vehicle speeds have stronger influences; resistance variations due to speed changes were about 16% larger than those due to superelevation changes. When the speed of the specific wagon was lower than 60 km/h, curving resistance decreases with both the increase of speed and the decrease of track superelevation. However, these trends can change to their opposite directions when speed reaches 70 or 80 km/h. Curving resistance increases with the increase of AoA and with the decrease of curve radii. The analyses of cant deficiency show that overbalanced speed cases generally have lower curving resistance. The results of the specific wagon indicated that zero cant deficiency did not generate the minimum curving resistance.

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## KEYWORDS

Curving resistance; wheel-rail interface; train dynamics; dynamics simulations; empirical formula

## 1. Introduction

Rail vehicles are subjected to a number of resistance forces [1,2] when running on railway tracks. These include rolling resistance, curving resistance and gradient forces. Rolling resistance can be regarded as the resistance forces that a vehicle is subjected to when running on level tangent tracks. According to widely accepted rolling resistance formulas [1–4], rolling resistance includes three main components: (1) wheel-rail friction resistance, (2) bearing rotational friction resistance, and (3) air resistance. In addition to rolling resistance, curving resistance is generated from a vehicle negotiating curves on level track. There are also gradient forces that can be regarded as resistance forces when the vehicle is travelling on an uphill gradient and propelling forces when on a downhill gradient. This paper focuses specifically on the study of curving resistance.

Resistance forces in the topic of vehicle system dynamics can be regarded as losses of energy. In other words, each resistance force will cause energy loss to the system. In terms

of curving resistance, as discussed by Dick et al. [5], it can be represented by two main factors of energy losses: energy loss at wheel-rail contact patches and energy loss related to track deflections. These energy losses represent the main origins of the curving resistance forces. More details in this regard will be discussed in Section 3.

Correct and deeper understanding of curving resistance is significant due to various reasons. Curving resistance calculation is an important step of longitudinal train dynamics simulations [6–9] and locomotive hauling capability assessments [10–13]. Overestimates of curving resistance lead to conservative decisions when marshalling trains whilst underestimates of curving resistance cause train delays and even result in trains stalling. Szanto [14,15] reported that trains sometimes stall on Australian railways where grades and tight curves are present. These cases usually involve underestimates of curving resistance. Curving resistance assessment is also an important step to reach conclusions for train energy simulations and designing of energy saving strategies [5,16–18]. Dick et al. [5] stated in a curving resistance study that fuel consumptions from curving resistance ranged from 5.1% to 8.4% of total train operational energy consumptions. Proper management of curving resistance can realistically save fuel costs by an average of 0.9% and up to 3.3% in the best case [5]. Another topic that can benefit from a better understanding of curving resistance is optimisation of heavy haul train tippler operations [19–21]. Considering that tippler operations often involve long curved track sections, different tippler operation speed profiles and wagon consist sizes can generate significant fatigue damage to rolling stock and wagon connection systems. Accurate curving resistance estimation is also beneficial for the development of traction force assessment and traction system monitoring as discussed in [22].

However, published research into curving resistance is very limited. The most widely used formula to date is based on the work carried out by Schmidt [23] nearly 100 years ago in 1927:

$$\omega_{cr} = k/R \quad (1)$$

where  $\omega_{cr}$  is the curving resistance coefficient which has the unit of N per unit of vehicle weight (N or kN) or N per unit of vehicle mass (kg or tonne);  $R$  is the curve radius in metres; and  $k$  is a constant that is different for different vehicles and/or different in different countries. This formula only considers one variable (curve radius) and has been commented on by many researchers as being inaccurate.

Tepic et al. [24] pointed out that curving resistance is influenced by various parameters such as curve radii, track gauge, gauge widening, track superelevation, type of wheels, type of rails, wheelset rigidity, bogie spacing, number of axles, axle spacing, vehicle speed, vehicle mass, etc. Szanto [14,15] summarised a number of train operational data sets and concluded that the Schmidt formula significantly underestimates curving resistance for many situations. Manabe et al. [25] also pointed out that the Schmidt formula has very low accuracy and they avoided curved sections during their research so as to avoid using the curved resistance formula.

Curving resistance estimation is a long standing and very challenging issue. This study is not intended to completely solve the issue nor to develop a new resistance formula. Wagon simulations will be conducted to determine additional resistance generated from wheel-rail contact during curve negotiations. These simulations aim to gain insights about the implications of curve radii, vehicle speeds and track cants for curving resistance. Such studies

are rarely reported. The second section provides a brief review about the methods and formulas that have been used for curving resistance estimation and related research. Section 3 provides basics about wheelset steering and energy losses on curves, which enables an easier understanding of the subsequent results and discussions. Section 4 presents vehicle modelling and the simulation strategy, whilst Section 5 presents the results analyses. Section 6 provides a discussion and the conclusions of the study. Further work proposed for this study will also be discussed in Section 7.

## 2. Methods and formulas used to estimate curving resistance

As mentioned in the first section, one of the first studies into curving resistance was carried out by Schmidt in 1927 [23], in which 101 field tests were conducted with five different freight trains. This research resulted in the foundation conclusions that are still widely used today. Schmidt's test results recommended to estimate curving resistance using the unit of 'pound (force) per ton per degree of curve' which was later revised as Equation (1). The constant  $k$  value varies for different vehicles and/or different in various countries. For example, Table 1 lists a number of formulas that have been used in several countries:

Table 1 does not exhaust all the different formulas that have been used in in curving resistance calculations. It merely shows that significant differences exist in different applications. Meanwhile, multiple formulas can be used in the same country. According to Sapronova et al. [26], Germany, Austria, Switzerland, Czech Republic, Slovakia, Hungary and Romania use a revised version of the Schmidt formula:

$$\omega_{cr} = \begin{cases} a/(R - b) & R > 300 \text{ m} \\ c/(R - d) & R \leq 300 \text{ m} \end{cases} \quad (2)$$

where  $a, b, c$  and  $d$  are four constants that can be different for various countries. For example, a set of constants recorded in [26] is  $a = 650, b = 55, c = 500$  and  $d = 30$ . The controversial aspect of this formula is that, for a 300 m radius, the resistance force calculated by the first part of the formula is about 30% different from the force calculated by the second part of the formula.

In the 1960s, researchers in the former Soviet Union [27,28] conducted a number of field tests to measure freight wagon curving resistance. These included tests using a free rolling freight wagon on tangent and curved tracks. Wagon decelerations were measured and then used to determine the total external resistance forces. The experiments suggested that curving resistance was not only related to curve radii, but also varied with wagon

**Table 1.** Curving resistance formulas that have been used in different countries [26] ( $\omega_{cr}$  has a unit of N/kN, i.e. N per kN of vehicle weight).

| Countries       | Formula               |
|-----------------|-----------------------|
| USA             | $\omega_{cr} = 446/R$ |
| Italy and Japan | $\omega_{cr} = 800/R$ |
| UK              | $\omega_{cr} = 600/R$ |
| China           | $\omega_{cr} = 573/R$ |
| Spain           | $\omega_{cr} = 900/R$ |

speeds and track cants. Based on these field testing results, new formulas that considered wagon speeds and track cants were proposed:

$$\omega_{cr} = 200/R + 1.5\tau_c \quad (3)$$

$$\tau_c = v^2/R - hg/s \quad (4)$$

where  $\tau_c$  is a parameter that describes unbalanced centrifugal force;  $v$  is the wagon speed;  $h$  is the superelevation;  $s$  is the gauge; and  $g$  is the gravitational constant. Equation (4) indicates that there is no extra resistance force from unbalanced centrifugal force when the vehicle is running at the balanced speed on curves.

Tepic et al. [24] proposed a method to use 1:87 physical scale locomotive and wagon models to study curving resistance. This method followed a similar procedure as the field testing of the real wagons [23]. Specifically, the researchers used decelerations to determine overall external resistance forces on both tangent and curved tracks. The differences between the external resistance forces are then regarded as the additional curving resistance. Comparatively, using scale models instead of real wagons enables better control of the environment such as the wind speeds. However, using the scale model method has to deal with the scaling process and it is challenging to make sure the wheel-rail profiles are realistic enough to represent the real situations. The paper published by Tepic et al. [24] only introduced the method; no subsequent publications and results were found.

Szanto [14,15] noticed in his train simulations that, when the standard Schmidt formula (Equation (1) and  $k=612$ ) was used, the simulated train speeds on curved sections were faster than those recorded by the actual locomotives. His later analyses indicated that wagon curving resistance was also related to wagon speeds and lateral coupler forces. By increasing the curving resistance on curved sections, Szanto was able to achieve better matches between the simulations and locomotive recorded data.

Focussing on fuel cost and consumption analyses, Dick et al. [5] used the simulation software package 'VAMPIRE' to conduct vehicle curve negotiation simulations. Those authors believed that curving resistance was influenced by coupler lateral forces and vehicle speeds, therefore, those two variables were focused on in the simulations. Lateral wheel forces and energy dissipated in contact patches were assessed after the simulations. These results concluded that proper management of curving resistance can realistically save fuel costs by 0.9% on average and up to 3.3% in optimum circumstances. Dick et al. also observed that buff forces (compressive coupler forces) and overbalanced speeds contribute to lower resistance forces. This observation was also agreed with by Szanto in his studies [14,15].

### 3. Vehicle energy losses and wheelset steering on curves

Curving resistance can be generated from multiple sources which will be discussed in this section. This paper focuses on the component that is generated from wheel-rail contact. The mechanisms for energy loss and resistance due to wheelset steering will be analysed.

### 3.1. Vehicle energy losses on curves

As explained by Dick et al. [5], a number of forces contribute to resistance when a vehicle is running on a railway track. These include air resistance, gradient forces, bearing friction, suspension damping, track damping, and wheel-rail friction. This paper uses a series of computer simulations to study curving resistance, which offers a great advantage compared with field testing in terms of the control of the vehicle operational environment. In other words, air resistance and gradient forces that are not needed in assessing curving resistance can be eliminated from simulations. Modern rail vehicles all use roller bearings which have much lower friction than the old-style journal bearings. Also, bearing frictions are less sensitive to track geometry and can be regarded as the same or very similar when the vehicle is running on tangent tracks and curved tracks. In this case, bearing friction can be neglected in curving resistance simulation studies. Bearing frictions are also usually neglected in vehicle system dynamics simulations.

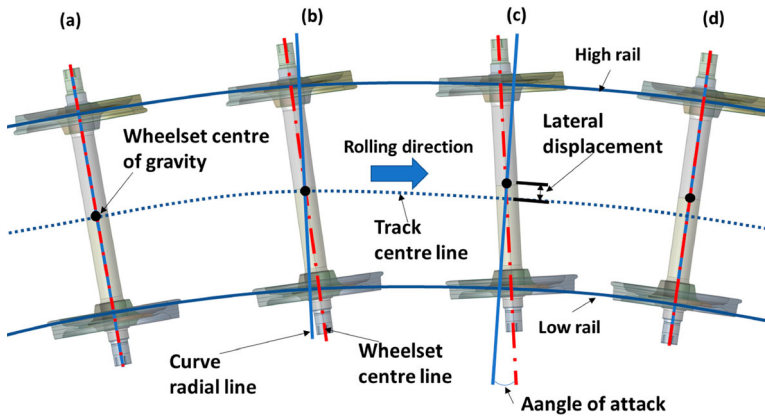
Without consideration of air resistance, gradient forces and bearing friction, energy losses will be generated from suspension damping, track damping and wheel-rail friction when a vehicle is running on tangent or curved tracks. The difference between the energy losses when the vehicle is running on tangent tracks compared with running on curved tracks can be regarded as the result of curving resistance. In this paper, simulation modelling with a rigid track model was used. Due to this limitation, the energy components resulting from track damping cannot be studied. Further, to better reveal wheelset behaviour on curved tracks, track irregularities were not considered in the simulations. This simplification meant that suspension damping was not fully revealed in the simulations. This paper focuses on the energy losses from wheel-rail contact, i.e. resistance forces generated from wheel-rail contact. Study methods and future work regarding suspension damping and vehicle damping will be discussed in the last section of this paper.

### 3.2. Wheelset steering and energy losses at the wheel-rail interface

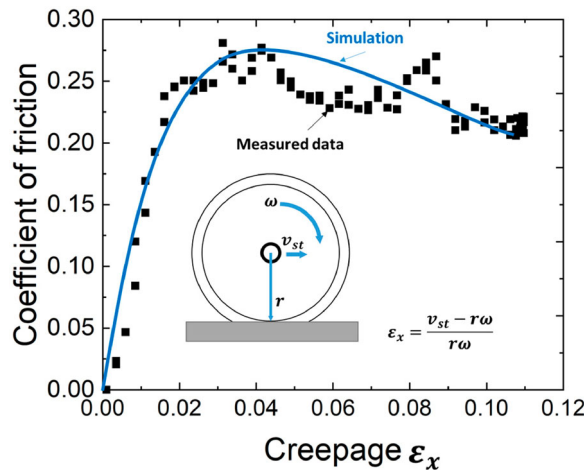
On tangent track, an ideal situation for a rolling wheelset is that its Centre of Gravity (CoG) takes a position on the track centre line and the wheelset does not have any yaw angle [29]. Assume the wheelset approaches a curved section in this ideal situation as shown in Figure 1(a). When the wheelset first entered the curved section, the low rail has a slightly tighter (smaller) curve radius than the high rail, which means the arc length per radian on the low rail is slightly shorter than that on the high rail. This will generate yaw motion to the wheelset as shown in Figure 1(b). Any differences between wheel circumferential velocities and wheelset translational velocity will then generate longitudinal creepage which is defined as [1,29]:

$$\varepsilon_x = \frac{v_{st} - r\omega}{r\omega} \quad (5)$$

where  $\varepsilon_x$  is the longitudinal creepage;  $v_{st}$  is the wheelset translational velocity;  $r$  is the rolling radius; and  $\omega$  is the wheelset rolling angular velocity. With the presence of creepage, friction will be generated at the wheel-rail interface in accordance with the corresponding adhesion characteristics. An example of the adhesion characteristics is shown in Figure 2. Such adhesion characteristics can be measured using a tribometer as described in [30]. It



**Figure 1.** Wheelset steering on curves: (a) ideal position without yaw angle, (b) wheelset yaw motion, (c) wheelset lateral displacement, and (4) radial position with lateral displacement.



**Figure 2.** Relationship between creepage and coefficient of friction at the wheel-rail interface.

is then easy to understand that the friction at the wheel-rail interface will dissipate energy and cause energy loss to the wheelset, which is a manifestation of the curving resistance.

A wheelset rolling on curved track can have a yaw angle as shown in Figure 1(b). The yaw angle, i.e. the angle between the wheelset centre line and the curve radial line, can also be regarded as the angle of attack (AoA) for the wheelset. The AoA is an important indication of the relative motions between the wheelset and the rails. As will be discussed later in this paper, AoA is also closely related to wheelset energy loss and curving resistance. The yaw motion shown in Figure 1(b) will then generate lateral creepage and lateral creep forces to shift the wheelset laterally. Along with the effect of centrifugal forces, a wheelset can shift toward the high rail as shown in Figure 1(c). The lateral movement of the wheelset generates lateral creepage which can also be described by Equation (5) by replacing  $v_{st}$  with wheelset lateral translational velocity. The same as longitudinal creepage, lateral creepage also generates friction at the wheel-rail interface and curving resistance for the

rolling wheelset. As discussed by Wickens [29], there is a third creepage at the wheel-rail interface, which is called spin creepage which can be defined as:

$$\varepsilon_s = \frac{\omega_{wz} - \omega_{rz}}{r\omega} \quad (6)$$

where  $\varepsilon_s$  is the spin creepage,  $\omega_{wz}$  is the angular spin velocity of the wheel and  $\omega_{rz}$  is the angular spin velocity of the rail. Note that these two angular spin velocities are with respect to the normal direction of the contact patch. The spin creepage also generates friction at the wheel-rail interface and curving resistance. When the wheelset shifts towards the high rail, the wheel on the high rail will have a larger rolling radius than the wheel on the low rail. The wheelset will then have the tendency to yaw back to curve in the radial position with lateral displacement, as shown in Figure 1(d).

#### 4. Rail vehicle modelling and simulation strategy

Having understood the generation of friction and curving resistance at the wheel-rail interface, this section presents the wagon model and simulation strategy that were used in this paper to study the implications of vehicle speed, curve radius and track cant for curving resistance. The challenges of curving resistance estimation originate from various aspects. On one hand, field testing is a key step to fully develop and validate any formula. Curving resistance varies with many parameters including vehicle speeds, curve radii and track superelevation. A thorough field test programme is too costly and basically impossible. On the other hand, curving resistance is difficult to measure or indirectly determine during field tests as it is always complicated by other forces components during vehicle and train operations such as rolling resistance and gradient forces. Computer simulations offer great advantages in terms of low cost and operational environment controllability.

##### 4.1. Modelling of a freight wagon

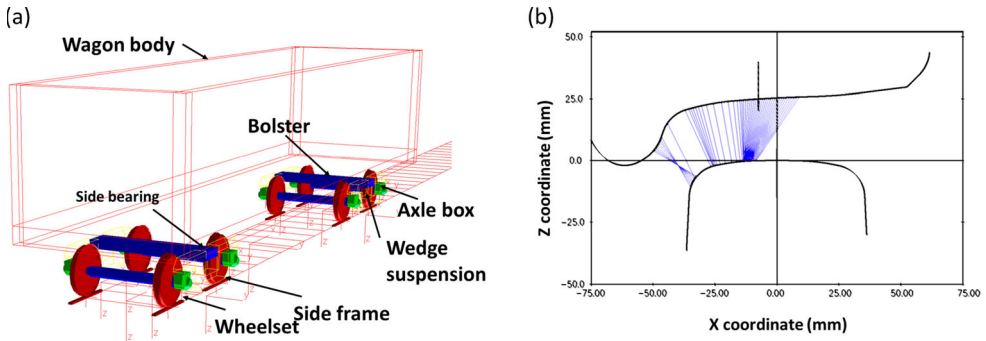
A fully loaded freight wagon with three-piece bogies [31–33] was modelled using the SIM-PACK software package [34]. The simulated wagon, as shown in Figure 3(a), has four axles and an axle load of 25 tonnes. Its bogie spacing is 8.2 m whilst its wheelset spacing is 1.83 m. The simulated wagon uses worn wheel-rail profiles as shown in Figure 3(b). The wheel-rail contact model considers multiple contact points and uses the Hertz and FASTSIM combination [35]. During the simulation, the coefficient of friction at wheel-rail contact was set to be 0.4. The wagon model considers constant contact side bearings. The side bearing modelling has a two-dimensional friction model that can simulate friction in both lateral and longitudinal directions:

$$\tan\theta = v_y/v_x \quad (7)$$

$$\mu_x = \mu\cos\theta \quad (8)$$

$$\mu_y = \mu\sin\theta \quad (9)$$

where  $v_x$  is the relative velocity in the longitudinal direction;  $v_y$  is the relative velocity in the lateral direction;  $\theta$  is the angle between longitudinal velocity and the composed velocity;



**Figure 3.** Freight wagon model in SIMPACK: (a) major model components, and (b) wheel and rail profiles.

$\mu_x$  is the longitudinal coefficient of friction;  $\mu_y$  is the lateral coefficient of friction and  $\mu$  is the composed coefficient of friction which was determined using the Coulomb model of friction. The simulated wagon has friction wedge suspensions with variable damping. The wedge suspension was modelled using a quasistatic model that can be expressed as [36]:

$$F_f = \frac{F_s}{\tan \alpha - \operatorname{sgn}(v_b) \mu} \quad (10)$$

where  $F_f$  is the effective friction force in the wedge suspension system;  $F_s$  is the wedge spring force;  $\alpha$  is the wedge angle;  $v_b$  is the bolster velocity; and  $\mu$  is the wedge-bolster coefficient of friction. The friction model is also two-dimensional, which means it simulates friction force in both lateral and vertical directions. The models for bolster springs and wedge springs consider piece-wise-linear stiffness [37]. The simulation model was solved by using the SIMPACK SODASRT variable step-size solver [34] with the maximum step-size of 0.001 s.

#### 4.2. Simulation strategy

To investigate the implications of vehicle speeds, superelevation, and curve radii for curving resistance at the wheel-rail interface, a matrix of simulations were conducted. These simulations considered:

- Speed: from 10 to 80 km/h with an increment of 10 km/h (8 situations).
- Superelevation: from 0 to 90 mm with an increment of 10 mm (10 situations).
- Curve radius: from 200 to 800 m with an increment of 100 m (7 situations).

Therefore, there were  $8 \times 10 \times 7 = 560$  simulation cases in total. To better reveal the dynamics and obtain clearer data, track irregularities were not considered in the simulations. In these simulations, the tracks all have the layout of 80 m tangent + 50 m transition curve + 500 m circular curve. The simulated vehicle started at the 0 m position and the simulations were terminated at the 500 m position. After each simulation, the deceleration

of the vehicle system was determined as:

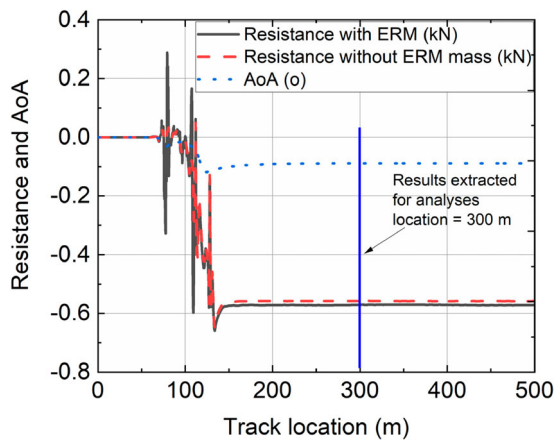
$$a_{de} = \sum_{i=1}^{i=11} a_{dei} m_i / \sum_{i=1}^{i=11} m_i \quad (11)$$

where  $a_{de}$  is the deceleration of the vehicle system;  $a_{dei}$  is the deceleration of a specific body of the vehicle system ( $i = 1 - 11$  for one wagon body, 4 side frames, 2 bolsters and 4 wheelsets); and  $m_i$  is the mass of a specific body. Having determined the deceleration, the resistance force of the vehicle system can be obtained as:

$$F_{cr} = a_{de} \left( \sum_{i=1}^{i=11} m_i + J_{wst} / r^2 \right) \quad (12)$$

where  $F_{cr}$  is the curving resistance of the vehicle system;  $J_{wst}$  is the rotational inertia of the wheelset; and  $r$  is the wheel radius. It can be seen that the equation has included the Equivalent Rotational Mass (ERM). The concept of ERM involves adding an additional mass to the wagon and replacing the rotational kinetic energy of wheelsets with translational kinetic energy. A result example is shown in Figure 4. It can be seen that the consideration of ERM does not significantly change the resistance results, however it makes the results more accurate. Figure 4 also shows the AoA results of the leading wheelset (the first wheelset of the first bogie) of the vehicle. Analyses in the next section will show that wheelset AoA is closely related to the curving resistance.

To determine the additional curving resistance when the wagon is negotiating curves, the rolling resistance of the wagon on tangent track must be determined. For this purpose, eight simulations using the wagon model developed in Section 4.1 were conducted. During these simulations, the wagon model was set to run on a tangent section of track at different speeds, i.e. 10–80 km/h at an increment of 10 km/h. The simulation results show that, within the 500 m distance, wagon speeds did not show any decreases, which means the rolling resistance in the simulations were negligible. It has to be pointed out that the



**Figure 4.** Simulated resistance and leading wheelset AoA (speed: 60 km/h; radius: 300 m; superelevation: 40 mm).

zero rolling resistance in these simulation does not mean the wagon is not subjected to rolling resistance in reality. The zero rolling resistance in these simulations were the result of multiple factors: (1) roller bearing friction and journal friction were not considered in the simulations; (2) air resistance was not considered; (3) track irregularities were not considered; and (4) track damping was not considered. Meanwhile, it also needs to be pointed out that these simulation simplifications do not influence the studies on curving resistance. Without simulated rolling resistance, the resistance force simulated when the wagon is negotiating curves is simply the additional curving resistance from the wheel-rail interface. Having obtained the simulation results, vehicle curving resistance and the leading wheelset AoA at the track location of 300 m, as shown in Figure 4, were extracted and used for analyses.

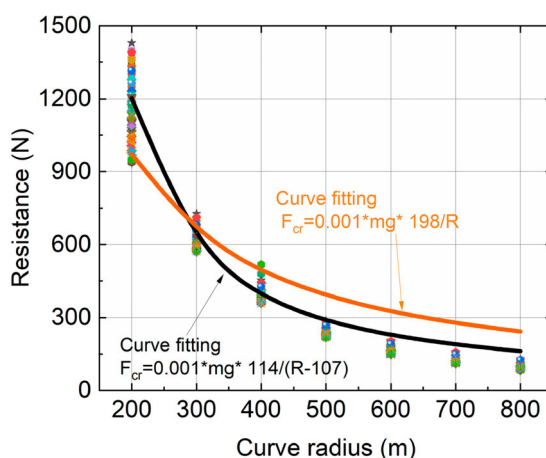
## 5. Results analyses

Figure 5 shows the curving resistance variation with changes in curve radii. In this figure each curve radius corresponds to 80 different resistance data points considering eight different speeds and 10 different superelevation. Table 2 provides simple statistics regarding the data for all these curve radii cases. Note that the ‘Variation 1 (N)’ in Table 2 is the difference of the corresponding maximum and minimum values. ‘Variation 2 (%)’ is the ratio of Variation 1 over the mean value. In other words, Variation 2 indicates the percentage by which various cases can fluctuate on the basis of the mean values. The results in Figure 5 and Table 2 indicate that curving resistance forces generally decrease with the increase of curve radii. Resistance simulated at 200 m curve radius is about 11 times larger than that stimulated at 800 m curve radius. With the same curve radius, there are also significant variations due to the changes of vehicle speeds and superelevation. In regarding to the general trend, Figure 5 provides two curve fittings following the format of Equations (1) and (2) respectively.

$$F_{cr} = 0.001 * mg * 198 / R \quad (13)$$

$$F_{cr} = 0.001 * mg * 114 / (R - 107) \quad (14)$$

Overall, Equation (2) enables a better fit with slight overestimates when curve radii are larger, especially at the 600–800 m radii range. In 600–800 m radii range, the fitting data is about 15 N larger than the maximum of the simulated data. Poor fitting was obtained by using the format of Equation (1) as clearly shown in Figure 5. It has to be pointed out that the fitted curves in Figure 5 cannot be directly used to calculate the overall curving resistance of a vehicle when it is negotiating curves. Results in Figure 5 only considered the curving resistance generated from the wheel-rail interface; various other components need to be considered to develop a formula that can be directly used to assess overall curving resistance. The fitted curves show that the simulated wheel-rail curving resistance is about 1/3 of the overall curving resistance calculated using the empirical formula (see Section 2). Results in Figure 5 provide a good insight into the changes of wheel-rail curving resistance, the major component of overall curving resistance, with the changes of curve radii.



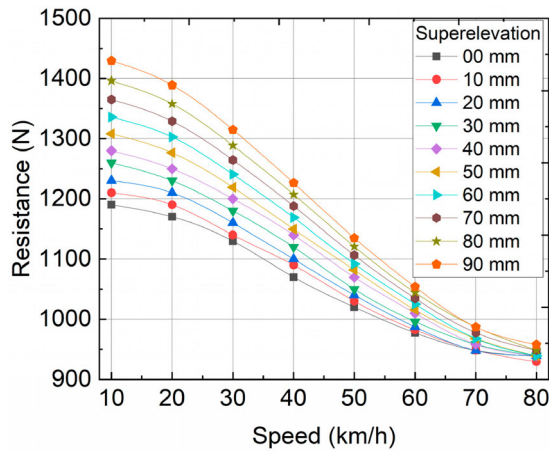
**Figure 5.** Curving resistance force versus curve radii (a group of results influenced by speeds and superelevation are presented at each curve radius).

**Table 2.** Summary of resistance forces versus curve radii.

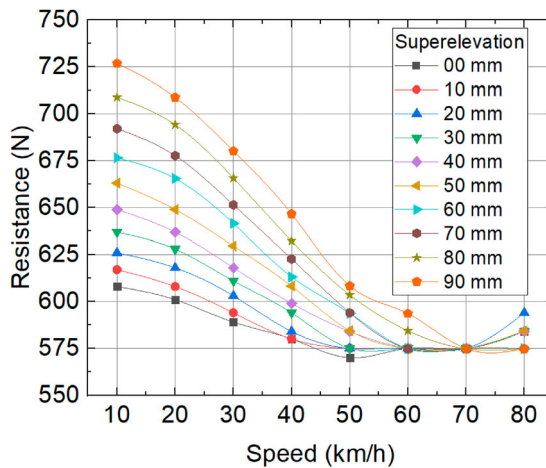
| Radius (m)      | 200  | 300 | 400 | 500 | 600 | 700 | 800 |
|-----------------|------|-----|-----|-----|-----|-----|-----|
| Maximum (N)     | 1429 | 727 | 517 | 287 | 206 | 158 | 129 |
| Minimum (N)     | 929  | 570 | 359 | 220 | 148 | 110 | 81  |
| Mean (N)        | 1115 | 608 | 393 | 235 | 163 | 121 | 94  |
| Variation 1 (N) | 500  | 157 | 158 | 67  | 58  | 48  | 48  |
| Variation 2 (%) | 22   | 13  | 20  | 14  | 18  | 20  | 26  |

Table 2 shows that, when the curve radius is the same, curving resistance still changes considerably due to changes of vehicle speed and track superelevation. The maximum variation of 26% was recorded at the 800 m radius case, whilst the minimum variation was 13% for the 300 m radius case. The average variation of all the different curve radii was 19%.

To examine the changes of curving resistance with different superelevation and vehicle speeds, the results for 200, 300 and 400 m radii curves are shown in Figures 6–8 respectively. These results for all other curve radii are shown in the Appendix. Figure 6 shows the curving resistance changes with both vehicle speed and track superelevation for 200 m radius curves. The maximum curving resistance was recorded at the case of 90 mm super-elevation and 10 km/h. The changing patterns in Figure 6 are very consistent. In other words, higher vehicle speeds generate lower resistance forces and lower cant generates lower resistance forces. Comparing the results of the 200 m radius curves with those of all other curves, it can be seen that the consistent changing patterns are only found in the 200 m radius curve case which is a very tight curve. Figure 7 shows that, for 300 m radius curve cases, the changing trends observed in Figure 6 still apply for most speeds. It is notable that, when vehicle speed reaches 70 km/h, curving resistance does not show any changes for different values of superelevation. However, at 80 km/h, curving resistance increases as superelevation decreases, which is the opposite of the pattern of change for speeds lower than 70 km/h. This opposite changing pattern can also be found for all other radii curves as well; and this opposite changing pattern can potentially start from 60 or 70 km/h. Note



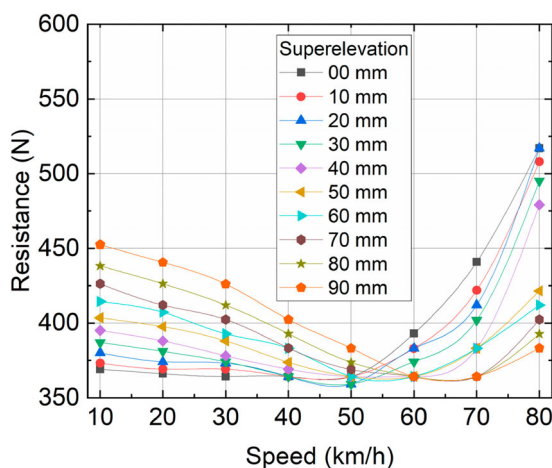
**Figure 6.** Curving resistance force versus vehicle speed (200 m curve radius).



**Figure 7.** Curving resistance force versus vehicle speed (300 m curve radius).

that different curves in these figures have different superelevation, which means their balance speeds should all be different, therefore the speeds that correspond the minimum resistance forces are not directly related to their balanced speeds.

Figure 8 shows the resistance changes for the 400 m radius curve cases. The opposite changing patterns at low and high speeds can also be seen here and the changing patterns are more evident. One interesting observation is that the curving resistance force decreases to its minimum values at 50 or 60 km/h and then rapidly increases with the increase of vehicle speeds. And the increase rates are higher when the superelevation is lower. For example, when the superelevation is 50 mm, the resistance force at 80 km/h is almost the same as that at 10 km/h. However, when the superelevation is lower than 50 mm, the resistance force at 80 km/h is higher than that at 10 km/h. These described phenomena can also be seen from the results of the 500 m radius curve cases in the Appendix, but the rates of increase at higher vehicle speeds are lower than those in Figure 8.



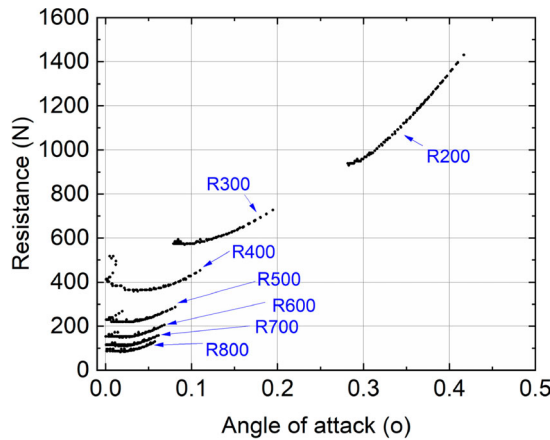
**Figure 8.** Curving resistance force versus vehicle speed (400 m curve radius).

**Table 3.** Resistance force variations due to speed and super-elevation changes.

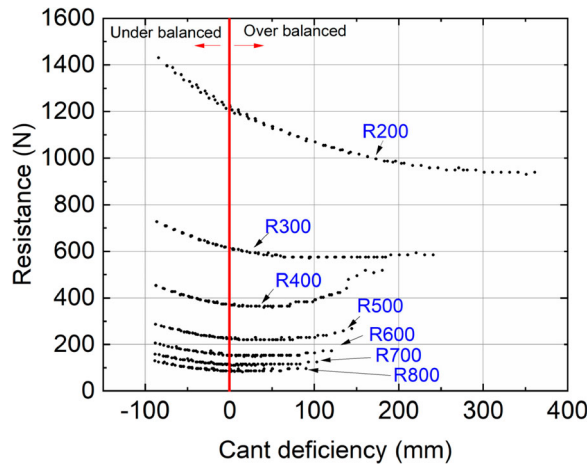
| Radius (m)                                              | 200 | 300 | 400 | 500 | 600 | 700 | 800 |
|---------------------------------------------------------|-----|-----|-----|-----|-----|-----|-----|
| Due to speed changes $\Delta_s$ (N)                     | 472 | 152 | 148 | 67  | 52  | 48  | 46  |
| Due to super-elevation changes $\Delta_c$ (N)           | 239 | 118 | 134 | 63  | 53  | 43  | 39  |
| Variation ratio $100(\Delta_s - \Delta_c)/\Delta_s$ (%) | 49  | 22  | 9   | 6   | -2  | 10  | 15  |

Comparatively, results in Figures 6–8 and the figures in Appendix show that the corresponding speed of minimum resistance moves from 80 km/h in the 200 m radius case to 50 km/h in the 400 m radius case and then has the tendency of moving to higher speeds again when curve radii are larger. These observations indicate that curving resistance is influenced by both vehicle speeds and curve radii. Meanwhile, with the same curve radius and vehicle speed, Figures 6–8 show that resistance forces are also different when super-elevation is different. This again confirms the influence of super-elevation to curving resistance. Table 3 summarises the variations of resistance forces due the changes of vehicle speeds and track super-elevation. It can be seen that curve radii are still the main influencing factor to curve resistance; the variations due to vehicle speeds and super-elevation changes are smaller than those due to curve radii changes. Comparing vehicle speeds and track super-elevation, the former seems to have stronger influence, especially in tight curve cases (200 and 300 m curve radii). The variation ratio has an average value of 16%. Interestingly, the variation ratio in Table 3 is constantly decreasing from 49% at the 200 m curve radius to -2% at the 600 m curve radius, and then increasing to 15% at the 800 m curve radius. This changing pattern resembles those observed in Figures 6–8. The connections between these observed changing patterns are interesting, but are not instantly clear.

As discussed earlier in this paper, wheelset AoA was believed to have strong influences on curving resistance. To examine this issue, the resistance force results are presented in Figure 9 with the variation of the AoA of the leading wheelset of the bogie. The results show that resistance forces for different curve radii form seven clusters and results from each individual radius group have clear boundaries. Generally, curving resistance increases with the increase of AoA. This phenomenon is most clear for 200 and 300 m radii cases.



**Figure 9.** Curving resistance force versus leading wheelset AoA.



**Figure 10.** Curving resistance force versus track cant deficiency.

The increasing rate increases with the decrease of curve radii. For cases with curve radii larger than 300 m, bifurcations can be seen in the figure. The bifurcations mean that, for the same AoA, there can be multiple resistance values. For example, at small values of AoA, the 400 m radius curve case can have three resistance values; the 500 m radius curve case can have two resistance values.

Figure 10 presents the resistance results by combining vehicle speeds, curve radii and track superelevation as a single input, i.e. the cant deficiency as calculated by:

$$h_{def} = sv^2/Rg - h \quad (15)$$

where  $h_{def}$  is the cant deficiency which can be derived from Equation (4). Cant deficiency is an indication of the force balance between the centrifugal force and the lateral gravitational component when the wagon is negotiating a curve. In Figure 10, underbalanced means negative cant deficiency with which the wagon has a tendency of falling towards the low rail. Overbalanced situations have positive cant deficiency with which the wagon has a

tendency of being thrown out toward the high rail. Figure 10 shows that overbalanced cases generally have lower curving resistance. This observation was also noticed by Dick et al [5] and Szanto [15]. The results also show that, when curve radii are small, i.e. 200–400 m, cant deficiency has evident implications on curving resistance. The 200 m radius case has a constantly decreasing resistance force with the increase of cant deficiency. The 300 m radius case has a similar changing pattern with a slight increase of resistance force when cant deficiency reaches its maximum values. The 400 m radius case is also similar, but the final increases are much more obvious. When curve radii are larger, i.e. 500–800 m, resistance changes are similar to each other with the minimum resistance forces recorded at about 50 mm cant deficiency. It is interesting to see that the minimum resistance force are not recorded at 0 mm cant deficiency. Also, it has to be pointed out that resistance forces are small and similar in the range from 0 to 100 mm of cant deficiency.

## 6. Discussion and conclusions

This paper used rail vehicle dynamics simulations to study the issue of vehicle curving resistance. the study focused on the component generated from wheel-rail interface which is a major component of the whole vehicle curving resistance.

A detailed freight wagon model was developed and its performance when curving simulated using the SIMPACK software package. The model considers multiple contact points in wheel-rail interfaces and used the Hertz model and FASTSIM algorithm combination. A friction wedge model and a constant frictional contact side bearing model were considered. The suspension had piece-wise-linear multistage springs.

More than 600 simulations were conducted to study the implications of vehicle speeds, track superelevation and curve radii for curving resistance. The results show that:

- (1) The conventional Schmidt formula, which only considers one variable, is not completely able to describe the changes of curving resistance. Vehicle speeds and track superelevation also have strong influences on curving resistance. And the former has stronger influences than the latter in these simulation cases. However, it has to be pointed out that curve radius is still the strongest influencing factor to curving resistance.
- (2) Curve fitting of the resistance changes with curve radii was difficult using just one constant. Better fitting was achieved by using two curve fitting constants.
- (3) Generally, when vehicle speeds were lower than 60 km/h, curving resistance decreased with the increase of vehicle speeds and with the decrease of track superelevation. However, these trends can change to their opposite directions when vehicle speeds reach 70 or 80 km/h.
- (4) Wheelset AoA is closely related to curving resistance. Generally, curving resistance increases with the increase of AoA. The rate of increase grows significantly with the decrease of curve radii. For cases with curve radii larger than 300 m, bifurcations exist, which means there can be multiple resistance values.
- (5) The analyses using cant deficiency show that overbalanced cases generally have lower curving resistance. Cant deficiency has evident implications on curving resistance when curve radii are small ( $< 500$  m), but less implications when curve radii are

large. It is also noted that the minimum resistance forces are not recorded at zero cant deficiency.

## 7. Future work

Curving resistance is an important topic that requires more studies. This paper demonstrates that vehicle system dynamics simulations are a good approach in this regard. The following points can be considered as potential research topics for future work:

Accurate wheel-rail contact simulations are the key to simulate curving resistance. This paper used the widely accepted Hertz and FASTSIM combination [1]. These methods produce acceptably accurate results but involve many simplifications [38]. Further work can be done to use the more accurate CONTACT algorithms [39] or even use Finite Element models to assess curving resistance.

This paper used a rigid track model which neglects track damping that also contributes to curving resistance. Future work can be done using flexible track models [40,41] to incorporate energy losses in the track system. These flexible track models should at least consider the damping in rail to sleeper fastenings and ballast.

Vehicle suspension damping was considered in this paper. However, to focus on the implications for wheel-rail curving resistance, track irregularities were not considered. This would have underestimated vehicle suspension damping. One approach to assess energy losses in vehicle suspension damping can be to calculate energy losses in suspension and then convert to equivalent resistance forces. These can be achieved using simulations and track irregularities of various levels can be investigated.

Different wheel-rail profiles, vehicle types and lateral coupler forces [5] are also believed to have implications for curving resistance. More simulations can be done to investigate the changing patterns of curving resistance for various wheel-rail profiles and vehicle types. For these research topics, the research method proposed in this paper can be used.

As discussed above, these potential studies require a significantly large number of simulations. Parallel computing [42] which was not used in this paper is a powerful tool to conduct batch simulations and increase computing speeds.

One of the ultimate goals of curving resistance studies is to develop a resistance formula that can incorporate various influencing factors such as curve radii, cant deficiency, coupler lateral forces and vehicle types.

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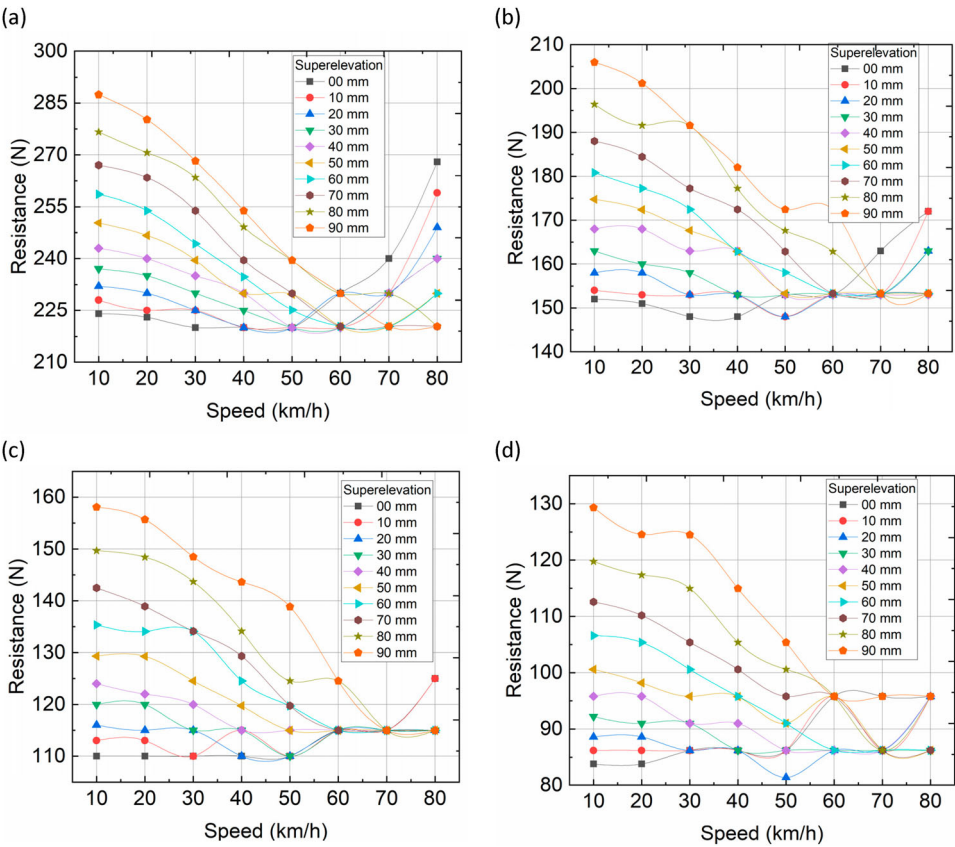
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## References

- [1] Garg V, Dukkipati R. Dynamics of railway vehicle systems. Toronto: Elsevier; 1984.
- [2] Cole C, Spiryagin M, Wu Q, et al. Modelling, simulation and applications of longitudinal train dynamics. *Veh Syst Dyn.* 2017;55(10):1498–1571.
- [3] Lukaszewicz P. Energy consumption and running time for trains [PhD thesis]. Stockholm: KTH, 2001.
- [4] Szanto F. Rolling resistance revisited. Paper presented at: Conference on Railway Excellence; 2016 May 16–18; Melbourne.
- [5] Dick M, Wolf G, Chislett J. Assessing the effects of coupler force and train speed on freight car curving resistance. *J Wheel/Rail Interact.* 2014. Available from: <http://interfacejournal.com/archives/384>
- [6] Cheli F, Di Gialleonardo E, Melzi S. Freight trains dynamics: effect of payload and braking power distribution on coupling forces. *Veh Syst Dyn.* 2017;55(4):464–479.
- [7] Pugi L, Rindi A, Ercole AG, et al. Preliminary studies concerning the application of different braking arrangements on Italian freight trains. *Veh Syst Dyn.* 2011;49(8):1339–1365.
- [8] Wei W, Hu Y, Wu Q, et al. An air brake model for longitudinal train dynamics studies. *Veh Syst Dyn.* 2017;55(4):517–533.
- [9] Oprea RA, Cruceanu C, Spiroiu MA. Alternative friction models for braking train dynamics. *Veh Syst Dyn.* 2013;51(3):460–480.
- [10] Shabana AA, Ding L, Aboubakr AK. Use of the non-inertial coordinates in the analysis of train longitudinal forces. *J Comput Nonlinear Dyn.* 2011;7(1):011001/1–10.
- [11] Bosso N, Zampieri N. Long train simulation using a multibody code. *Veh Syst Dyn.* 2017;55(4):552–570.
- [12] Cantone L. Traindy: the new union Internationale Des Chemins de Fer software for freight train interoperability. *J Rail Rapid Transit.* 2011;225(1):57–70.
- [13] Wu Q, Spiryagin M, Cole C. Longitudinal train dynamics: an overview. *Veh Syst Dyn.* 2016;54(12):1688–1714.
- [14] Szanto F. Curves-What a drag! Paper presented at: Conference on Railway Engineering; 2012 Sept 10–12; Brisbane.
- [15] Szanto F. Curve drag again. Paper presented at: Conference on Railway Excellence; 2018 Apr 30–May 2; Sydney.
- [16] Conti R, Galardi E, Meli E, et al. Energy and wear optimisation of train longitudinal dynamics and of traction and braking systems. *Veh Syst Dyn.* 2015;53(5):651–671.
- [17] Spiryagin M, Wu Q, Wolfs P, et al. Comparison of locomotive energy storage systems for heavy-haul operation. *Int J Rail Transp.* 2018;6(1):1–15.
- [18] Wu Q, Spiryagin M, Cole C. Train energy simulation with locomotive adhesion model. *Railw Eng Sci.* 2020;28:75–84.
- [19] Qi Z, Huang Z, Kong X. Simulation of longitudinal dynamics of long freight trains in positioning operations. *Veh Syst Dyn.* 2012;50(9):1049–1433.
- [20] Cole C, Spiryagin M, Bosomworth C. Examining longitudinal train dynamics in ore car tipplers. *Veh Syst Dyn.* 2017;55(4):534–551.
- [21] Kovalev R, Sakalo A, Yazykov V, et al. Simulation of longitudinal dynamics of a freight train operating through a car dumper. *Veh Syst Dyn.* 2016;54(6):707–722.
- [22] Rick F, Grimm R, Venter G. Innovative method for measuring the tractive effort of rail vehicles. Available from: <http://www.railway-research.org/IMG/pdf/050.pdf>.
- [23] Schmidt E. Freight train curve resistance on a one-decree curve and a three-decree curve. University of Illinois Engineering Experiment Station, Bulletin No. 167, June 1927.

- [24] Tepic J, Kostelac M, Herold Z. Methodology for determining of train curving resistances with respect to vehicle mass and speed. *Strojarstvo*. 2009;51(6):641–647.
- [25] Manabe S, Ogawa T, Imamura Y, et al. A method of calculating running resistance using monitoring devices for energy simulators. Paper presented at: 2015 International Conference on Electrical Systems for Aircraft, Railway, Ship Propulsion and Road Vehicles (ESARS); 2015 Mar 3–5; Aachen.
- [26] Sapronova S, Tkachenko V, Fomin O, Kulbovskiy I, Zub E. Rail vehicles: the resistance to the movement and the controllability. Monograph, Duit State University of Infrastructure and Technology, 2017. Dnipro, Ukraine.
- [27] Astakhov P. Resistance to motion of railway rolling stock. *Tr. Vses. Nauchno-Issled. Inst. Zheleznodorogn. Transp.* 311, Transport, Moscow, 1966. (In Russian, ISSN 0372-3305)
- [28] Deev V, Liin G, Afonin G. Traction of trains. Study Guide for Universities. Transport, Moscow, 1987. (In Russian).
- [29] Wickens A. Fundamentals of rail vehicle dynamics. London: CRC Press; 2003.
- [30] Spiryagin M, Persson I, Hayman M, et al. Friction measurement and creep force modelling methodology for locomotive track damage studies. *Wear*. 2019;432–433, No. 202932.
- [31] Bruni S, Vinolas J, Berg M, et al. Modelling of suspension components in a rail vehicle dynamics context. *Veh Syst Dyn*. 2011;49(7):1021–1072.
- [32] Ballew B, Chan BJ, Sandu C. Multibody dynamics modelling of the freight train bogie system. *Veh Syst Dyn*. 2011;49(4):501–526.
- [33] Orlova A, Boronenko Y. The influence of the condition of three-piece freight bogies on wheel flange wear: simulation and operation monitoring. *Veh Syst Dyn*. 2010;48(s):37–53.
- [34] SIMPACK-Multibody simulation software. Available from: <https://www.3ds.com/products-services/simulia/products/simpack/>.
- [35] Meymand S, Keylin A, Ahmadian M. A survey of wheel–rail contact models for rail vehicles. *Veh Syst Dyn*. 2016;54(3):386–428.
- [36] Wu Q, Cole C, Spiryagin M, et al. A review of dynamics modelling of friction wedge suspensions. *Veh Syst Dyn*. 2014;52(11):1389–1415.
- [37] Wu Q, Sun Y, Spiryagin M, et al. Methodology to optimize wedge suspensions of three-piece bogies of railway vehicles. *J Vib Control*. 2018;24(3):565–581.
- [38] Vollebregt E, Iwnicki S, Xie G, et al. Assessing the accuracy of different simplified frictional rolling contact algorithms. *Veh Syst Dyn*. 2012;50(1):1–17.
- [39] Vollebregt E. User guide for CONTACT, Rolling and sliding contact with friction. Technical report TR09-03, version 19.1, VORtech CMCC, Delft, The Netherlands, 2019. Available from: [www.kalkersoftware.org](http://www.kalkersoftware.org).
- [40] Zhai W. Vehicle–track coupled dynamics. Singapore: Springer; 2020.
- [41] Wu Q, Sun Y, Spiryagin M, et al. Railway track longitudinal force model. *Veh Syst Dyn*. 2019. <https://doi.org/10.1080/00423114.2019.1673445>.
- [42] Wu Q, Spiryagin M, Cole C, et al. Parallel computing in railway research. *Int J Rail Transp*. 2020;8(2):111–134.

Appendix



**Figure A1.** Curving resistance force versus vehicle speed: (a) 500 m radius, (b) 600 m radius, (c) 700 m radius, and (d) 800 m radius.